

Gemini Chat

Visualizin...nb - Colab

 PDF

In my prompt, I pass in the notebook PDF and ask it the corresponding questions. Since it cannot see all visualizations, I compare its predicted results to my own observations.

Help me solve the conceptual questions from the attached deep learning notebook PDF that I already ran. We will go one question at a time.

Let's start with these first 2 questions:

(a) Attention in GPT: Run part a of the notebook and generate the corresponding visualizations

- i. What similarities and differences do you notice in the visualizations between the examples in this part? Explore the queries, keys, and values to identify any interesting patterns associated with the attention mechanism.
- ii. How does attention differ between the different layers of the GPT model? Do you notice that the tokens are attending to different tokens as we go through the layers of the network?

✦ Based on the "Visualizing BERT" notebook PDF you provided and standard behavior of Transformer models, here are the answers to your conceptual questions regarding Part (a).

(a) Attention in GPT

i. What similarities and differences do you notice in the visualizations between the examples in this part?

Gemini's analysis about early layers from what it knows about transformers matches what I observe in the notebook. It also correctly mentions masked attention and references specific examples from the notebook, showing that it actually read the uploaded context and not just spitting out generic attention visualization answers.

- **Similarity: The Autoregressive (Causal) Constraint**

The most critical similarity across all examples ("The dog ran," "The dog sitting...", and "Jaime Salazar...") is that tokens **never attend to future tokens**. GPT is an autoregressive model trained to predict the next word, so it is strictly prevented from "seeing" what comes next.

- In your PDF, look at the visualization for "The dog ran". The word "dog" attends to "The" and "dog", but it does *not* attend to "ran".
- Similarly, in "The dog sitting in the car", the word "sitting" connects to "The", "dog", and "sitting", but has no connection lines to "in", "the", or "car".
- This creates a distinctive "triangle" or "look-back only" pattern in the attention visualization.

- **Similarity: Heavy Local Attention** In Layer 0 (which is what is displayed in your screenshots), you likely see a

strong diagonal line. Tokens pay a lot of attention to themselves and the word immediately preceding them. This is because the first layer often processes local, n-gram style relationships.

- **Difference: Context Length** As the sentences get longer (e.g., the "Jaime Salazar" example), the *potential* history a token can attend to grows, but the masking constraint remains identical.

ii. How does attention differ between the different layers of the GPT model?

Since you have run the interactive notebook, you can toggle the **Layer** dropdown. Here is what you will observe as you move from Layer 0 to deeper layers:

- **Layer 0 (Surface Level):** As seen in your PDF, this layer is very **local and positional**. Tokens mostly attend to the

My observations
in the notebook
match what
Gemini predicted
in later layers.

current position or the immediate previous position. It is gathering raw information about the neighboring words.

- **Middle/Deep Layers (Semantic Level):** As you move to higher layers (e.g., Layer 5 or 10), the attention patterns usually become more **sparse and long-range**.
 - Instead of just looking at the previous word, a token might look back 10 words to find a specific reference.
 - For example, in the sentence "Jaime Salazar is running... Mr.", a deeper layer might show "Mr." attending strongly to "Salazar" or "Jaime" to resolve the entity, ignoring the filler words in between.
- **Conclusion:** Yes, tokens attend to **different tokens** as you go through the layers. The model moves from processing "which word is next to me?" (Layer 0) to "which previous words give me meaning and context?" (Deep Layers). However, the causal constraint (never looking forward) applies to **all** layers.

Would you like to move on to Part (b) regarding BERT's bidirectional attention?

(b) BERT pays attention: Run part b of the notebook and generate the corresponding visualizations

- i. Look at different layers of the BERT model in the visualizations of part (b) and identify different patterns associated with the attention mechanism. Explore the queries, keys, and values to further inform your answer. For instance, do you notice that any particular type of tokens are attended to at a given timestep?
- ii. Do you spot any differences between how attention works in GPT vs. BERT? Think about how the model architectures are different.

iii. For the example with syntactically similar but definitionally different sentences, look through the different layers of the two BERT networks associated with sentence a and sentence b, and take a look at the queries, keys, and values associated with the different tokens. Do you notice any differences in the embeddings learned for the two sentences that are essentially identical in structure but different in meaning?

iv. For the pre-training related examples, do you notice BERT's bi-directionality in play? Do you think pre-training the BERT helped it learn better representations?

For Part (b), Gemini correctly makes the distinction between masked attention in GPT vs BERT. It also talks about the SEP and CLS tokens which I didn't expect, and it was right about the dense cross-attention visualizations as well which I did see.

Here are the answers to the conceptual questions for Part (b), drawing from the visualizations in your PDF and the architectural differences between BERT and GPT.

(b) BERT pays attention

i. Patterns in BERT Layers and Token Attention

In the visualizations (specifically on **Pages 4 and 5**), you can observe distinct patterns that differ from GPT:

- **The [SEP] and [CLS] Attractors:** A very common pattern in BERT (often visible in deeper layers or specific heads) is that many tokens will heavily attend to the [SEP] (separator) or [CLS] (classification) tokens.
 - This often looks like "vertical bars" in the visualization where every word connects to the special tokens.
 - **Why?** The model uses these tokens as "no-op" or "global context" aggregators. When a specific word doesn't find a strong relationship with other words, it

"parks" its attention on [SEP] or [CLS] to preserve the signal without distorting the meaning.

- **Dense Cross-Attention:** Unlike GPT, you will see tokens attending to words that appear *after* them in the sentence. This creates a dense web of lines rather than the clean "triangle" seen in GPT.

ii. Differences between GPT and BERT Attention

The most striking difference is visible when comparing the plots on **Page 2 (GPT)** vs. **Page 4 (BERT)**:

- **GPT (Unidirectional/Causal):** As noted in part (a), GPT's attention is a **triangle**. Tokens can only see themselves and the past (left context). The lines never reach "forward" to the right.
- **BERT (Bidirectional):** BERT's attention is a **full square** (conceptually). Tokens attend to the entire sequence simultaneously. In the visualization, this means you see lines connecting a word (e.g., "party") to words that come both before *and* after it (e.g., "wanted" and "bought").
- **Architecture:** This stems from their training objectives. GPT is a **Decoder** trained to predict the next token (hence the mask). BERT is an **Encoder** trained to fill in blanks ([MASK]) and predict sentence relationships, requiring access to the full context on both sides.

iii. The "Play" Homonym Example (Syntactically Similar Sentences)

Referencing the visualizations on **Page 5** for:

1. "I was going to **play** at the park" (Verb)
 2. "I was going to a **play** in the park" (Noun)
- **Attention Differences:** Yes, the attention mechanism distinguishes these meanings.

Gemini also correctly analyzed how the word "play" had major differences in attention between the 2 sentences. The attention aggregated different context words

- In the **Verb** case ("to play"), the token "play" likely attends strongly to "going" and "to" to establish the future tense action.
- In the **Noun** case ("a play"), the token "play" likely attends to the determiner "a" and the preposition "in", grounding it as an object/event rather than an action.
- **Embeddings:** Because the *values* (context vectors) are aggregated based on these different attention weights, the final embedding for the word "play" is mathematically different in these two sentences. This allows BERT to solve tasks like Word Sense Disambiguation dynamically.

iv. Pre-training and Bidirectionality (Reversed/Scrambled inputs)

Referencing the scrambled examples on **Page 6** (e.g., "party a have to wanted I"):

Gemini's explanation was pretty educational regarding BERT's bidirectionality allowing it to not break when order changes due to pre-training.

- **Bidirectionality in Play:** Even when the sentence is reversed, BERT's attention lines still connect related words (like "party" and "cake") regardless of their position. Because BERT is bidirectional, it doesn't "break" merely because the order is flipped; it can still "see" the keywords across the sequence.
- **Did Pre-training Help?** Yes. BERT was pre-trained on the **Next Sentence Prediction (NSP)** task.
 - In the example where Sentence B is "*I bought a cake for it*", BERT's attention heads (specifically in the deeper layers) likely show strong connections between "party" (Sentence A) and "cake" (Sentence B).
 - This shows the model learned a high-level representation of **cause and effect** or **theme consistency** during pre-training. It "knows" that

cakes and parties belong together, allowing it to link these sentences even if the syntax is slightly garbled.

Would you like to continue to Part (c) and discuss BERT's multiple attention heads?

(c) BERT has multiple heads!: Run part c of the notebook and generate the corresponding visualizations.

i. Do you notice different features being learned throughout the different attention heads of BERT? Why do you think this might be?

ii. Can you identify any of the different features that the different attention heads are focusing on?

In this response, Gemini definitely uses some knowledge about attention visualization in general as it mentions vocab like "diffuse mesh" which wasn't in the notebook. However, it is educational and its analysis aligns exactly with what I saw in the notebook. It was able to look at the individual attention plots on the grid, which is also pretty impressive CV at play

Based on the visualizations in Part (c) of the notebook (specifically the grid of attention heads shown on **Page 8**), here are the answers to your questions:

(c) BERT has multiple heads!

i. Do you notice different features being learned throughout the different attention heads of BERT? Why do you think this might be?

Yes, distinct patterns are clearly visible across the different heads. Looking at the grid on Page 8, you can see that the attention "texture" varies significantly from square to square (where each square represents a specific Layer-Head combination):

- Some heads show clear **diagonal lines**.
- Some heads show **vertical stripes** (attending heavily to one specific token).
- Some heads look like a **diffuse mesh** (broad attention).

Why this happens: This is the fundamental purpose of **Multi-Head Attention**. It allows the model to focus on different types of information simultaneously at different "subspaces."

- If BERT had only one attention head per layer, it would have to average conflicting signals (e.g., looking at the previous word for syntax *and* looking at a distant word for semantic reference). This would create a muddy, uninformative vector.
- By having multiple heads, one head can specialize in **grammar** (syntax), another in **meaning** (semantics), and another in **position** (structure), allowing the model to build a rich, multifaceted representation of the text.

Gemini's prediction about seeing diagonals does align with what I observed in the notebook. It makes interesting analogies to n-grams and convolutional filters which is also insightful.

ii. Can you identify any of the different features that the different attention heads are focusing on?

Based on the grid patterns visible in the PDF, here are the specific features the heads correspond to:

1. Positional/Local Attention (Diagonals):

- **Visual:** You will see strong diagonal lines in the grid (often in earlier layers).
- **Feature:** This head is focusing on **local context**—specifically the words immediately preceding or following the current token (n-grams). It mimics the behavior of a convolutional filter, capturing immediate phrases.

2. No-Op / Global Context (Vertical Stripes):

- **Visual:** You will see solid vertical bars where almost every word connects to a single position (usually the first or last column).
- **Feature:** This represents attention to the **special tokens** `[CLS]` (start) or `[SEP]` (end).

Gemini explains concepts not mentioned in the notebook like "no-ops" although I would need to do further reading to see if that's actually true or not.

- **Function:** When a head can't find a useful meaningful connection for a word, it often "parks" its attention on these special tokens. This serves as a "no-op" (doing nothing) or aggregates a global sentence representation.

3. Syntactic/Semantic Relations (Sparse/Specific Lines):

- **Visual:** Irregular, specific lines connecting different parts of the sentence (not just diagonals or stripes).
- **Feature:** These heads are tracking linguistic dependencies. For example, a head might be responsible for linking **verbs to their direct objects** (e.g., "wanted" -> "party") or **pronouns to their antecedents**. This is where the model captures the logic of the sentence.

(d) Visualizing untrained attention weights

- i. What differences do you notice in the attention patterns between the randomly initialized and trained BERT models?
- ii. What are some words or tokens that you would expect strong attention between? What might you guess about the gradients of this attention head for those words?

✦ Based on the experiment setup in the notebook (Part d) and the general principles of the BERT model shown in the previous sections, here are the answers to your questions regarding untrained attention weights.

(d) Visualizing untrained attention weights

i. What differences do you notice in the attention patterns between the randomly initialized and trained BERT models?

- **Trained Model (Structured):** As seen in Part (c), the trained model displays distinct, recognizable patterns. You see clear **diagonal lines** (attention to local context), **vertical stripes** (attention to `[SEP]` or `[CLS]` tokens), and sparse, specific connections between linguistically related words. The attention is focused and sharp.
- **Untrained Model (Random/Diffuse):** When the weights are randomly initialized (using the `init_weights` function shown in the code), these structures disappear. The attention patterns appear **chaotic, random, or uniformly diffuse**. There are no diagonals or vertical attractors because the model has not yet learned the positional encodings or the special role of `[SEP]` tokens. The lines connecting tokens would look like a messy web with no discernible logic.

ii. What are some words or tokens that you would expect strong attention between? What might you guess about the gradients of this attention head for those words?

For the sentences *"I am happy"* and *"I am not sad"*:

- **Expected Strong Attention:** You would expect strong attention between words that modify or relate to each other semantically. Specifically:
 - **"not" and "sad":** The word "not" is crucial for negating "sad." To understand the sentiment or predict the word "sad," the model must attend heavily to "not."
 - **"happy" and "sad":** These are antonyms contrasting the state of the speaker in the two sentences.
- **Gradients:**

For this last question, Gemini is slightly off from what I observed as i saw more equal attention rather than chaotic lines, but otherwise it's response is very informative and doesn't just tell me what the gradients would be like, but also "why" because of backprop, which i found to be insightful.

- In the **untrained model**, the attention score between "not" and "sad" is likely random and low (close to zero or average).
- However, because this relationship is essential for minimizing the loss (e.g., determining the sentence meaning or predicting masked words), the **gradients** for the attention weights connecting "not" and "sad" would be **large and positive**.
- The backpropagation process is "screaming" at the model to increase the connection strength between these specific words, driving the weights away from their random initialization toward the structured patterns seen in the trained model.